FINAL AGENDA

CHiMaD Phase Field Methods XVIII

April 22 – 24, 2025

Hybrid via Zoom & @CHiMaD HQ

All times in CST

Tuesday, April 22

12:00PM – 01:00PM Welcome & Lunch

01:00PM – 01:40PM **Spatially Resolved Modeling of Lithium Dendrite Growth**
Edwin Garcia, *Purdue University*

01:40PM – 02:20PM **Phase-Field Simulation of Electrodeposition and Stripping of Li Metal Batteries**
Ye Cao, *University of Texas at Arlington*

02:20PM – 03:00PM Group Photo & Break

03:00PM – 03:40PM **A Continuum Electro-chemo-mechanical Gradient Theory Coupled with Phase-field Fracture: On Modeling the Growth of Li-filaments in All-solid-state Batteries**
Donald Bistri, *University of Illinois Urbana-Champaign*

03:40PM – 04:20PM **Quantitative Phase Field Model for Electrochemical Systems**
Jin Zhang, *Northwestern University*

04:20PM – *open ended* Discussion: **Phase Field Methods for Electrodeposition / Benchmarks**
Leads: Jin Zhang, *Northwestern University* / Jon Guyer, *NIST*

05:30PM Dinner

Wednesday, April 23

08:30AM – 09:00AM Breakfast

09:00AM – 09:40AM **Learning Phase-Field Models from Image Data**
Martin Bazant, *Massachusetts Institute of Technology* (Virtual)

09:40AM – 10:10PM Lightning Talks (for speakers in Europe and Asia, late submissions are also accepted)
Consecutive Intra-particle Phase Transitions in LiFePO4 Battery Electrodes
Souzan Hammadi, *Uppsala University* – Sweden (Virtual)

10:10AM – 10:15AM **Welcome by new NAISE director**
Neelesh Patankar, *Northwestern University*

10:15AM – 10:40AM Break

10:40AM – 11:20AM **Elucidating the Interplay between Thermodynamics, Kinetics, and Electrochemistry in Battery Cathode Materials**
Katsuya Thornton, *University of Michigan, Ann Arbor*

11:20AM – 12:00PM **Battery Material and Microstructure Design with the Multi-phasefield Method**
Simon Daubner, *Imperial College London – UK* (Virtual)

12:00PM – 01:00PM Lunch

01:00PM – 01:40PM **Optimization of Electrode Architecture for Phase-separating Materials Guided by High-throughput Style Electrochemical Simulations**
Hui-Chia Yu, *Michigan State University* (Virtual)

01:40PM – 02:10PM **PFHub: Updates**
Daniel Wheeler, *NIST*

02:10PM – 03:00PM Discussion: **Recommended Practices for Phase Field Modeling**
Lead: David Montiel, *University of Michigan, Ann Arbor* (TBC)

03:00PM – 03:30PM Break & Snacks

03:30PM – *open ended*

Lightning Talks (Open to additional talks identified during the meeting)

Using Compile-time Symbolic Algebra to Build CUDA Kernels for Phase-field Equations
Steven Silber, *Western University – Canada*

Phase Field Modeling of a Dendrite Growth and Dead Lithium Formation in Lithium Anode and Mitigation Using a Protective Layer.
Bharat Pant, *The University of Texas at Arlington*

Model-based Design of Coatings to Modulate the Kinetics of Phase Separating Battery Particles and Minimize Phase Separation
Samuel Degnan-Morgenstern, *Massachusetts Institute of Technology*

AMMBER: The AI-enabled Microstructure Model BuildER - Status Updates
Xander Mensah, *University of Michigan, Ann Arbor*

Differentiable Programming/Physics for Phase Field Model Parameter Optimization
Andrew B Li, *University of Michigan, Ann Arbor*

PRISMS-MultiPhysics – An Open-Source Coupled Phase Field-Crystal Plasticity Framework and its Application to Simulate Twinning
David Montiel, *University of Michigan, Ann Arbor* (Virtual)

An Overview of Performance Improvements in PRISMS-PF 3.0
Jason Landini, *University of Michigan, Ann Arbor*

Simulation of Stack Pressure Induced Stress at the Li/SE Interface Using the Smoothed-boundary Method
Jonathan Gjerde, *University of Michigan, Ann Arbor*

Thursday, April 24

08:30AM – 09:00AM Breakfast

09:00AM – 9:40AM **Image-based Damage Modeling of Electrode Microstructures in Lithium-ion Batteries**
Adam Boyce, *University College Dublin – UK* (Virtual)

09:40AM – 10:20AM **Chemo-mechanical Coupling in Phase-field Methods: Benchmarks and Comparison**
Thea Kannenberg, *KIT – Germany* (Virtual)

10:20AM – 10:30AM Break

10:30AM – 12:00PM Discussion: **Next Steps & Community Action Items**
Lead: Jon Guyer, *NIST*

12:00PM Adjourn & Lunch to-go